

COURSE SYLLABUS

Administering a SQL Database (2025)

COURSE NUMBER

65316A

DURATION

5 days · 16 modules

AUDIENCE LEVEL

Intermediate

FORMAT

Instructor-Led / Self-Paced

Course syllabus

COURSE OVERVIEW

This five-day instructor-led course provides students with the knowledge and skills to administer a SQL Server 2025 database infrastructure. Students will learn to deploy and configure SQL Server 2025 instances, manage database objects, implement security, monitor performance, and automate administrative tasks using SQL Server Management Studio 22, PowerShell, and command-line tools.

AT COURSE COMPLETION

After completing this course, you will be able to:

Configure and deploy SQL Server 2025 instances using SQL Server Management Studio 22, PowerShell, and command-line tools

Administer database objects, implement comprehensive security models, and manage user permissions effectively

Monitor database performance, optimize query execution, and implement proactive maintenance strategies

Automate administrative tasks using SQL Server Agent, PowerShell scripts, and command-line utilities

Implement comprehensive backup, recovery, and disaster recovery strategies to protect business-critical data

PREREQUISITES

- Basic knowledge of the Microsoft Windows operating system
 - Working knowledge of Transact-SQL
 - Working knowledge of relational databases
 - Working knowledge of older editions of SQL Server Management Studio
 - Experience with database design
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COURSE OUTLINE

MODULE 1 SQL Server Security

129 min

This module covers the fundamentals of SQL Server security, including authentication modes, login creation, and security principal management using SQL Server Management Studio 22.

LESSONS

- Lesson 1:** SQL Server Security Architecture Overview
- Lesson 2:** Authentication Modes: Windows vs. Mixed Mode
- Lesson 3:** Creating and Managing Logins in SSMS 22
- Lesson 4:** Password Policies and Expiration Settings
- Lesson 5:** Security Principals: Logins, Users, and Roles
- Lesson 6:** Credential Management and Best Practices

LAB EXERCISES

- Configuring SQL Server Authentication and Security Principals

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Configure Windows and SQL Server authentication modes using SSMS 22
- Create and manage SQL Server logins and security principals
- Implement password policies and security best practices
- Evaluate security implications of different authentication methods

This module explores fixed server roles, fixed database roles, and custom role creation to implement role-based access control in SQL Server 2025.

LESSONS

Lesson 1: Understanding Role-Based Access Control in SQL Server

Lesson 2: Fixed Server Roles and Their Permissions

Lesson 3: Fixed Database Roles in SQL Server 2025

Lesson 4: Creating and Managing Custom Database Roles

Lesson 5: Role Membership Management and Best Practices

Lesson 6: Application Roles and Advanced Role Scenarios

LAB EXERCISES

Configuring Server and Database Roles for Multi-Tier Application Security

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Identify and assign fixed server roles using SSMS 22

Configure fixed database roles and custom database roles

Implement role-based security using least-privilege principles

Manage role membership and permissions effectively

This module focuses on granular permission management, including granting, denying, and revoking permissions on database objects and schemas.

LESSONS

Lesson 1: Understanding SQL Server
Permission Architecture

Lesson 2: GRANT, DENY, and REVOKE:
Permission Control
Statements

Lesson 3: Schema-Level Permissions
and Access Control

Lesson 4: Object-Level Permissions for
Granular Control

Lesson 5: Permission Inheritance and
Ownership Chains

Lesson 6: Troubleshooting Effective
Permissions

LAB EXERCISES

Implementing Schema and Object
Permissions in SQL Server 2025

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Grant, deny, and revoke permissions on database objects

Configure schema-level and object-level permissions

Implement effective permission hierarchies

Troubleshoot permission-related access issues using SSMS 22

MODULE 4 Encrypting Data with TDE and Always Encrypted

124 min

This module covers data encryption strategies including Transparent Data Encryption (TDE) and Always Encrypted features in SQL Server 2025.

LESSONS

- Lesson 1:** Data Encryption Fundamentals and Security Strategies
- Lesson 2:** Transparent Data Encryption Architecture and Implementation
- Lesson 3:** TDE Key Management and Certificate Operations
- Lesson 4:** Always Encrypted Concepts and Architecture
- Lesson 5:** Configuring Always Encrypted in SSMS 22
- Lesson 6:** Encryption Performance Considerations and Best Practices

LAB EXERCISES

- Implementing TDE and Always Encrypted in SQL Server 2025

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Implement Transparent Data Encryption (TDE) for database files
- Configure Always Encrypted for column-level encryption using SSMS 22
- Manage encryption keys and certificates
- Evaluate encryption performance and security implications

This module introduces SQL Server auditing capabilities, including server and database audit specifications for compliance and security monitoring.

LESSONS

- Lesson 1:** Understanding SQL Server Auditing Architecture
- Lesson 2:** Configuring Audit Targets and Destinations
- Lesson 3:** Creating Server Audit Specifications
- Lesson 4:** Implementing Database Audit Specifications
- Lesson 5:** Querying and Analyzing Audit Logs
- Lesson 6:** Audit Performance and Storage Management

LAB EXERCISES

- Configuring and Analyzing SQL Server Auditing

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Create and configure SQL Server audit objects
- Implement server audit specifications for instance-level events
- Configure database audit specifications for data access tracking
- Review and analyze audit logs using SSMS 22 and T-SQL

MODULE 6 Advanced Auditing and Compliance Features

144 min

This module explores advanced auditing capabilities including fine-grained auditing, dynamic data masking, and compliance automation using SQL Server 2025 features.

LESSONS

Lesson 1: Advanced Compliance and Auditing Overview

Lesson 2: Fine-Grained Audit Action Specification

Lesson 3: Dynamic Data Masking Implementation

Lesson 4: Row-Level Security with Auditing Integration

Lesson 5: Custom Audit Policies and Retention Strategies

Lesson 6: Compliance Reporting and Monitoring Integration

LAB EXERCISES

Implementing Fine-Grained Auditing and Dynamic Data Masking

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Implement fine-grained auditing for specific user actions
- Configure dynamic data masking for sensitive data protection
- Create custom audit policies using T-SQL and PowerShell
- Automate compliance reporting using audit data

This module covers SQL Server recovery models and their impact on backup and restore operations, along with backup strategy design principles.

LESSONS

Lesson 1: Understanding SQL Server Recovery Models

Lesson 2: Simple Recovery Model: Characteristics and Use Cases

Lesson 3: Full Recovery Model and Point-in-Time Recovery

Lesson 4: Bulk-Logged Recovery Model: Balancing Performance and Protection

Lesson 5: Backup Types and Strategies

Lesson 6: Recovery Objectives: RTO and RPO

LAB EXERCISES

Configuring Recovery Models and Implementing Backup Strategies

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Evaluate recovery models: simple, full, and bulk-logged

Configure database recovery models using SSMS 22

Design backup strategies aligned with recovery requirements

Calculate recovery objectives and backup schedules

This module provides hands-on training in performing full, differential, and transaction log backups using SSMS 22, T-SQL, and PowerShell.

LESSONS

Lesson 1: Understanding SQL Server Backup Architecture

Lesson 2: Full Database Backup Operations

Lesson 3: Differential Backup Implementation

Lesson 4: Transaction Log Backup Operations

Lesson 5: Backup Compression and Encryption

Lesson 6: Backup Verification and Validation Techniques

LAB EXERCISES

Implementing a Comprehensive SQL Server Backup Strategy

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Perform full database backups using SSMS 22 and T-SQL

Execute differential backups and understand their dependencies

Create transaction log backups for point-in-time recovery

Verify backup integrity using RESTORE VERIFYONLY and checksums

This module teaches database restore operations including full, differential, and point-in-time recovery scenarios using SQL Server 2025 and SSMS 22.

LESSONS

- Lesson 1:** Understanding Database Restore Operations
- Lesson 2:** Full Database Restore Procedures
- Lesson 3:** Differential and Transaction Log Restore Sequences
- Lesson 4:** Point-in-Time Recovery and STOPAT Options
- Lesson 5:** Restoring to Alternate Locations and Database Renaming
- Lesson 6:** Restore Verification, Validation, and Troubleshooting

LAB EXERCISES

- Restoring SQL Server Databases with Multiple Recovery Scenarios

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Restore full database backups to original and alternate locations
- Perform differential restore operations
- Execute point-in-time recovery using transaction log backups
- Troubleshoot common restore errors and issues

MODULE 10 Importing and Exporting Data

120 min

This module covers data import and export techniques using SQL Server Integration Services tools, bcp utility, and SSMS 22 import/export wizards.

LESSONS

Lesson 1: Overview of Data Import and Export in SQL Server 2025

Lesson 2: SQL Server Import and Export Wizard in SSMS 22

Lesson 3: bcp Utility for High-Performance Bulk Operations

Lesson 4: BULK INSERT Statement for T-SQL Integration

Lesson 5: OPENROWSET for Ad-Hoc Data Access

Lesson 6: Format Files for Complex Import Scenarios

LAB EXERCISES

Bulk Data Operations with SQL Server Import/Export Tools

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Import data using SQL Server Import and Export Wizard in SSMS 22

Export data to various file formats using SSMS 22 tools

Use bcp utility for high-performance bulk operations

Implement BULK INSERT and OPENROWSET for programmatic data import

MODULE 11 Automating SQL Server Management with SQL Server Agent

134 min

This module teaches automation using SQL Server Agent, including job creation, step configuration, and schedule management in SSMS 22.

LESSONS

- Lesson 1:** SQL Server Agent
Architecture and Capabilities
- Lesson 2:** Creating Jobs in SSMS 22
- Lesson 3:** Job Step Types and
Configuration
- Lesson 4:** Job Schedules: Time-Based
and Event-Based Automation
- Lesson 5:** Job History, Logging, and
Troubleshooting
- Lesson 6:** Multi-Server Management
and Job Security

LAB EXERCISES

- Creating and Managing Automated
Database Maintenance Jobs

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Create and configure SQL Server Agent jobs using SSMS 22
- Define job steps with T-SQL, PowerShell, and operating system commands
- Implement job schedules with recurring and event-driven triggers
- Monitor job execution history and troubleshoot failures

MODULE 12 Managing Alerts and Notifications

124 min

This module covers SQL Server Agent alerts, operator configuration, and notification methods including email and pager notifications.

LESSONS

- Lesson 1:** Alerts and Notifications
Architecture Overview
- Lesson 2:** Alert Types and Triggering
Conditions
- Lesson 3:** Creating and Configuring
Alerts
- Lesson 4:** Operators and Notification
Schedules
- Lesson 5:** Database Mail Configuration
- Lesson 6:** Alert Response Automation
and Best Practices

LAB EXERCISES

- Configure SQL Server Alerts and Email
Notifications

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Create SQL Server Agent alerts for errors and performance conditions
- Configure operators and notification delivery methods
- Implement Database Mail for email notifications
- Design notification escalation and response procedures

This module teaches SQL Server administration using PowerShell, including the SqlServer module, cmdlets, and scripted automation techniques.

LESSONS

- Lesson 1:** Introduction to PowerShell for SQL Server
- Lesson 2:** Installing and Configuring the SqlServer Module
- Lesson 3:** Connecting to SQL Server with PowerShell
- Lesson 4:** Executing Queries with Invoke-Sqlcmd
- Lesson 5:** Automating Backups with PowerShell
- Lesson 6:** Working with SQL Server Management Objects (SMO)
- Lesson 7:** Error Handling and Logging in PowerShell Scripts
- Lesson 8:** Scripting Repetitive Tasks and Batch Operations

LAB EXERCISES

- Automating SQL Server Administration with PowerShell

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Connect to SQL Server instances using PowerShell SqlServer module
- Execute T-SQL queries using Invoke-Sqlcmd cmdlet
- Automate database administration tasks with PowerShell scripts
- Manage SQL Server configuration using PowerShell cmdlets

MODULE 14 Monitoring SQL Server with Extended Events and DMVs

134 min

This module covers performance monitoring using Extended Events and Dynamic Management Views (DMVs) to diagnose and troubleshoot SQL Server issues.

LESSONS

Lesson 1: Introduction to SQL Server Monitoring

Lesson 2: Extended Events Architecture and Benefits

Lesson 3: Creating Extended Events Sessions in SSMS 22

Lesson 4: Understanding Dynamic Management Views

Lesson 5: Identifying Blocking and Deadlocks with DMVs

Lesson 6: Query Performance Analysis Using DMVs

Lesson 7: Proactive Monitoring and Alerting with DMV Queries

LAB EXERCISES

Monitoring SQL Server Performance with Extended Events and DMVs

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Create and configure Extended Events sessions using SSMS 22

Query Dynamic Management Views to identify performance bottlenecks

Analyze wait statistics and resource consumption patterns

Implement proactive monitoring strategies using DMVs

This module provides practical troubleshooting techniques for common SQL Server problems including connectivity, performance, and corruption issues.

LESSONS

Lesson 1: SQL Server Troubleshooting Methodology

Lesson 2: Diagnosing Connectivity and Authentication Issues

Lesson 3: Identifying and Resolving Blocking

Lesson 4: Analyzing and Preventing Deadlocks

Lesson 5: Troubleshooting Query Performance Problems

Lesson 6: Detecting and Repairing Database Corruption

LAB EXERCISES

Diagnosing and Resolving SQL Server Performance and Connectivity Issues

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Diagnose and resolve connectivity and authentication issues

Troubleshoot blocking and deadlock situations

Identify and resolve slow query performance problems

Detect and repair database corruption using DBCC commands

MODULE 16 SQL Server 2025: New Features and Enhancements (Optional)

143 min

This optional module explores new features and enhancements in SQL Server 2025, with emphasis on AI integration, SSMS 22 capabilities, JSON support, and T-SQL improvements.

LESSONS

Lesson 1: SQL Server 2025 Innovation Landscape

Lesson 2: AI and Machine Learning Integration

Lesson 3: SSMS 22 New Features and Enhancements

Lesson 4: Enhanced JSON Support and Querying

Lesson 5: T-SQL Language Enhancements and New Functions

Lesson 6: Performance and Scalability Improvements

Lesson 7: Security and Compliance Features

Lesson 8: Cloud Integration and Hybrid Scenarios

LAB EXERCISES

Exploring AI Integration and Enhanced T-SQL Features in SQL Server 2025

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Identify new AI and machine learning features in SQL Server 2025

Utilize SSMS 22 enhancements for improved productivity

Implement new JSON and T-SQL functionality

Distinguish between new features and upgraded capabilities from prior versions

BOOTCAMP MODE — CAPSTONE LABS

One integrative Capstone Lab is generated per course day. Each capstone presents a new scenario requiring students to independently apply the combined skills from that day's modules — no procedure, no hints. Students determine their own approach. Capstone Labs are the afternoon activity in Bootcamp delivery and replace the lecture portion of standard Instructor-Led Training.

DAY 1 CAPSTONE Modules 1, 2, 3

SKILLS TO BE INTEGRATED

- Configure authentication modes and create SQL Server logins using SSMS 22
- Implement password policies and manage login properties through PowerShell and GUI
- Create and assign server-level and database-level roles to security principals
- Manage role membership and permissions using T-SQL and SSMS 22
- Grant, deny, and revoke object-level permissions on database schemas and objects
- Configure user mapping to databases and assign database roles
- Implement least-privilege access control using custom database roles

DAY 2 CAPSTONE Modules 4, 5, 6

SKILLS TO BE INTEGRATED

- Enable Transparent Data Encryption on user databases and manage encryption keys
- Configure Always Encrypted for column-level encryption using SSMS 22 wizard
- Manage certificate and key stores for database encryption
- Create and configure server audit specifications for security events
- Implement database audit specifications to track data access and modifications
- Configure audit targets and review audit logs using SSMS 22 and T-SQL
- Implement fine-grained auditing for compliance requirements
- Configure dynamic data masking for sensitive column protection
- Create and manage custom audit policies using T-SQL and PowerShell

DAY 3 CAPSTONE Modules 7, 8, 9, 10

SKILLS TO BE INTEGRATED

- Configure database recovery models (simple, full, bulk-logged) using SSMS 22 and T-SQL
- Perform full database backups using SSMS 22 backup wizard and T-SQL commands
- Execute differential and transaction log backups on schedule
- Verify backup integrity using RESTORE VERIFYONLY and checksum options
- Restore full database backups to original and alternate locations
- Perform point-in-time recovery using transaction log backups
- Restore differential backups and apply transaction log sequences
- Import data using SQL Server Import and Export Wizard in SSMS 22
- Export database objects and data using bcp utility and SSMS 22 tools
- Use BULK INSERT and OPENROWSET for high-volume data operations

DAY 4 CAPSTONE Modules 11, 12, 13

SKILLS TO BE INTEGRATED

- Create and schedule SQL Server Agent jobs using SSMS 22 interface
- Configure job steps with T-SQL, PowerShell, and operating system commands
- Implement job schedules with recurring and event-driven triggers
- Create SQL Server Agent alerts based on error severity and performance conditions
- Configure operators and notification methods for alerts and job failures
- Implement email notifications using Database Mail configuration
- Connect to SQL Server instances using PowerShell SqlServer module
- Execute T-SQL queries and retrieve results using Invoke-Sqlcmd cmdlet
- Automate database maintenance tasks using PowerShell scripts and scheduled tasks
- Manage SQL Server configuration settings using PowerShell Set-SqlInstance cmdlets

DAY 5 CAPSTONE Modules 14, 15, 16

SKILLS TO BE INTEGRATED

- Create Extended Events sessions using SSMS 22 wizard to capture query performance
- Configure event targets and filters for focused performance monitoring
- Query Dynamic Management Views (DMVs) to identify resource bottlenecks
- Diagnose blocking and deadlock issues using sys.dm_exec_requests and related DMVs
- Troubleshoot slow query performance using execution plans and wait statistics
- Resolve connectivity issues using SQL Server error logs and network traces
- Identify and resolve database corruption using DBCC CHECKDB commands

Capstone content is embedded in the student manuals and instructor guides — one section per day, after the final module for that day. A standalone Capstone Slides (.pptx) file is also included in the Exclusive download package.